Weightage

The coursework is 30% of the assessment for the module (groups of 2-3)

APU Campus Wide Drinks and Snack Ordering Kiosk System

Asia Pacific University has recently moved into a brand new campus which has been purpose built to addressed the needs of students and staff. In order to provide additional convenience the University has decided to implement a kiosk system specifically to allow users to order drinks and snacks. The system is intended to alleviate unnecessary queues around a soon to open snack booth which will be located on the 3rd floor. Initially five kiosks are planned, although depending on the success of the system there are plans to introduce a further five.

The kiosks allow the user (it is anticipated that the primary users will be students although staff/lecturers may also use the system) to browse a menu of drinks and snacks, with the option of limited cutomisation. For instance, the user may choose to order a standard Kopi Ais (Ice Coffee), available in 3 sizes, or opt to customise the drink by adding extra sugar/less sugar, more milk, specifying the quantity of ice etc. Similarly a variety of other coffee based beverages can be modified to include toppings (cream, sprinkles, chocolate sauce etc). The kiosks should also allow the user to order a limited variety of snacks (although this is likely to be increased at a later date), such as curry puffs, kuih, pau and a small selection of freshly made sandwiches. In the case of the latter, these too can be customised in terms of the ingredients (lettuce, onions, tuna, cheese, and the choice of dressing).

Payment is to be made via the Student/Staff AP card, in keeping with the cashless system which has been implemented across campus. Students thus swipe their ID cards at the kiosk and order their desired drink and/or snack, with the cost of order deducted from their credit/balance. The order is then sent to the preparation area of the main snack booth.

You are required to design and develop a system which provides the above functionality in such a way as to be **extensible** and **maintainable.** As there are five kiosks linked to the order notification system located at the preparation area of the snack booth, you are encourage to think about concurrency in your design. In particular, the system should be designed so as to allow changes, such as price alternations or alterations to availability, to be made centrally (from the snack booth's system) and to subsequently take effect across all corresponding order kiosks.

Summary of requirements

The functionality the system provides includes the following:

- Order Kiosk:
    - identify the user (who they are and the their current credit/balance)
    - view drinks/snacks
    - modify/customise drinks and snacks
    - order drink(s)/Snack

  - update user's card balance (deduct the cost of the order)
  - update product information (price, availability)
  - log transactions at each kiosk
- Admin-Staff System:
  - Order notification: What, by whom etc
  - ability to update product availability etc

(Plus any other functionality you think relevant!)

Whilst you are required to design a solution to address the above scenario and provide a skeletal implementation (see below), your primary goal is to not only design a system which addresses the above requirements but also adheres to good OO design principles - (relatively) easy to maintain, and extensible to meet the future needs of the business.

Assignment Requirements

a. Provide a high level Use Case Diagram of the above scenario, with supporting descriptions of the main use cases
b. Design a Class Diagram to represent the design of your system
c. Create Sequence Diagrams for 3 Use Cases
d. Provide an updated Class Diagram with 2-3 design patterns
e. Provide Sequence Diagrams to indicate the interaction of your patterns.
f. Implement your patterns in Java or C++
g. Write a brief evaluation of your solution and choice of patterns, as well as the suitability of each design pattern for implementing object oriented solutions.

Assignment Deliverables:

You are required to submit a report and code on the Friday of the last week of the module - (week 15). Your report should include the following:

- A cover page
- Table of contents
- Overview of the scenario
- Use Case Diagram (with a brief description)
- Class Diagram (no design patterns)
- Sequence Diagrams (with a brief description)
- Brief report describing the use of design patterns
- Refined class diagram with design patterns (supported by a brief description)
- Critical appraisal report
- References (if appropriate)